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SPECIAL SECTION

Indigenous co-stewardship of North American moose: recommendations and a vision for a restoration framework

Seth A. Moore[1] | William J. Severud[2] | Tiffany M. Wolf[3] | Katharine Pelican[4] | Joseph Bauerkemper[5] | Michelle Carstensen[6] | Steven K. Windels[7]

[1]Department of Biology and Environment, Grand Portage Band of Lake Superior Chippewa, 27 Store Road, Grand Portage, MN 55605, USA

[2]Department of Natural Resource Management, South Dakota State University, McFadden Biostress Laboratory 143, Box 2140B, Brookings, SD 57007, USA

[3]Veterinary Population Medicine, College of Veterinary Medicine, University of Minnesota-Twin Cities, Room 495 Animal Science/Veterinary Medicine Building, 1988 Fitch Avenue, St. Paul, MN 55108, USA

[4]Strategic Partnership and Research Collaborative, The Research and Innovation Office, University of Minnesota - Twin Cities, R285 Learning and Environmental Sciences Building, 1954 Buford Avenue, St Paul, MN 55108, USA

[5]Department of American Indian Studies, University of Minnesota Duluth, Cina Hall 110, 1123 University Drive, Duluth, MN 55812, USA

[6]Minnesota Department of Natural Resources, 5463 West Broadway, Forest Lake, MN 55025, USA

[7]Voyageurs National Park, 360 Hwy 11 E, International Falls, MN 56649, USA

Abstract

Moose (*Alces alces*; mooz [singular], moozoog [plural] in Anishinaabemowin, Ojibwe language) are an important species to many Indigenous rights-holders and stakeholders throughout their circumpolar range. Management of moose can often lead to conflict when various perspectives of Indigenous nations are not recognized or appreciated. During the 55th North American Moose Conference and Workshop held in Grand Portage, Minnesota, USA, we held a workshop with 145 participants centered around co-stewardship of moose among various Indigenous nations, federal, state, and provincial governments, academia, and non-governmental agencies. Using a facilitator, the participants identified opportunities and challenges surrounding issues related to moose management. Participants then further identified priority improvements and action steps for co-stewardship. Six core principles of Indigenous co-stewardship were developed: 1) recognition of Indigenous Sovereignty, which specifies that co-stewardship must begin with a recognition of the sovereignty of Indigenous nations and their inherent rights to manage, conserve, and preserve natural and cultural resources within their ancestral

Seth A. Moore and William J. Severud Co first and senior.

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Correspondence
William J. Severud, South Dakota State University, Department of Natural Resource Management, McFadden Biostress Lab, Brookings, SD 57007, USA.
Email: william.severud@sdstate.edu

Seth A. Moore, Department of Biology and Environment, Grand Portage Band of Lake Superior Chippewa, 27 Store Road, Grand Portage, MN 55605, USA.
Email: samoore@boreal.org

lands; 2) shared responsibility, where co-stewardship is a shared responsibility between Indigenous nations, federal governments, and state governments; 3) cultural and ecological respect, which stipulates that co-stewardship should honor the cultural significance of moose to Indigenous nations and recognize the ecological importance of moose within the broader ecosystem; 4) inclusive decision-making, which details that co-stewardship requires inclusive and equitable decision-making processes that involve meaningful consultation and consent from Indigenous nations; 5) resource sharing and capacity building, where co-stewardship involves the sharing of resources and knowledge between Indigenous and non-Indigenous partners; and 6) adaptive management and sustainability, specifying that co-stewardship should embrace adaptive management principles, where management strategies are continuously evaluated, adjusted, and improved based on new information and changing conditions. Some of the key takeaways from the workshop included that it will be essential to integrate Indigenous ways of knowing into an equitable and inclusive management system, there are existing models of co-stewardship that can be built upon, it is critical to build trust among all key stakeholders and rights-holders, and it will be important to establish formal and informal collaborative systems among all partners to support co-stewardship at all levels. We discuss a study and synthesis on Indigenous co-stewardship of moose and offer a synopsis and recommendations to advance restoration of moose in North America.

KEYWORDS
Alces alces, co-management, co-stewardship, Indigenous, moose, mooz, OH-SMART, policy

Indigenous societies have stewarded landscapes since prehistoric times, but the value systems and sustainable approaches historically used by those societies and nations have been impeded by colonization and a Western mindset focused on the idea that humans have dominion over, rather than existing within, the earth and its environment (Clauson 2015). In the treaty-signing era in the United States from 1768-1868, many Tribes, particularly those in the Midwest, reserved rights to hunt, fish, and gather in large territories otherwise ceded to the United States federal government (Pevar 2012). Under the United States constitutional framework, treaties are accorded status as the supreme law of the land, superseding all state and local policy and withstanding any contradictory federal policy that does not expressly address treaties (U.S. Constitution, Article VI, Supremacy Clause). Many treaties were signed prior to recognition of various state boundaries and Tribes now reside within the boundaries of states that were arbitrarily constructed around them. Usufructuary rights to use and benefit from land while the ownership of which belongs to another entity were reserved by Indigenous nations in

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many treaties. The usufructuary rights to hunt, fish, and gather ensured that vast areas of public land must remain available for subsistence practices of the sovereign Indigenous nations. Thus, those lands must have non-toxic air, water, and foods, and are to be co-managed by the Indigenous nations, states, and federal entities that have trust responsibilities to steward those lands. Even when usufructuary rights are acknowledged, the trend since Western colonization has been to private ownership over public ownership. Typically, treaty rights have not been acknowledged or allowed to be implemented on privately held lands—and thus the rights are limited from the outset. At a global governance level, the United Nations Declaration on the Rights of Indigenous Peoples recognizes the urgent need to respect and promote the inherent rights of Indigenous Peoples affirmed in treaties, agreements, and other constructive arrangements (United Nations General Assembly 2007). The United Nations further asserts a human right to a clean, healthy, and sustainable environment (United Nations General Assembly 2022*a*, *b*).

The concept of sovereignty entails culturally grounded, internally affirmed, and externally acknowledged authority, capacity, and liberty to determine governance. Stated more concisely, sovereignty is the authority of a people to govern themselves (i.e., self-governance). For Indigenous Peoples, sovereignty is inherent (Pevar 2012). Rather than deriving from external sources or being delegated to Indigenous Peoples by an external power, or even a constructed document like a constitution, Indigenous sovereignty is rooted from within. It arises by way of deep-history long-standing relationships between a people and their homelands, and between a people and other peoples, both human and non-human. Through deep history, unfolding across countless generations, collective peoples come to cohere as nations, and nations come to determine how they relate to themselves, their places, and other peoples—how they self-govern.

A vital contour of the exercise of sovereignty emphasizes diplomacy: relations with other peoples. Through diplomatic agreements with colonizing nations, Indigenous nations made sophisticated, difficult, heavily burdened decisions (often under staggering duress) granting limited permissions to colonizing nations to share and co-govern what became known as ceded territories. Territorial cessions made in United States–Indian treaties were not and are not absolute conveyance but rather were and are conditional and limited arrangements by which Indigenous nations preserved their continued inherent rights while authorizing colonizing nations to share and co-manage lands and resources (Deloria and Wilkins 2000, Kannan 2007, Pevar 2012, Bauerkemper 2016). Colonizing nations have failed to abide by these diplomatic relationships both through neglect of commitments made in treaties and through societal and governance behaviors that, despite reaching extremely far beyond the limited treaty permissions granted by Indigenous nations, have nevertheless been normalized. Returning with humility to treaty relations and reckoning these diplomacies as the sole and exclusive pathway for establishing legitimate non-Indigenous societies on this continent would fundamentally require cooperative resource stewardship, while also providing a model for understanding and configuring authority within co-stewardship practices (Bauerkemper 2016). Co-stewardship must first and foremost honor the primacy of inherent Indigenous sovereignty just as it flows from that sovereignty.

There have been calls to expand indigenization of conservation to include collaborations between non-Indigenous and Indigenous governments seen as integral to effective wildlife stewardship (Eichler and Baumeister 2018, Artelle et al. 2021, Hessami et al. 2021). Wildlife resources should be held in the public trust (Organ et al. 2012) and Hessami et al. (2021:1287) indigenized this tenet of the North American Model of Wildlife Conservation to include, "Wildlife are stewarded through an interwoven set of values and visions to holistically manage ecosystems. Indigenous and non-Indigenous governments are held accountable by the public to effectively do so."

Moose (*Alces alces*; mooz [singular], moozoog [plural] in Anishinaabemowin, Ojibwe language) are a vital subsistence food for many Indigenous nations in the northern hemisphere (Popp et al. 2019, Wolf et al. 2021, Priadka et al. 2022, Severud et al. 2022). In Minnesota, USA, moose have been the primary terrestrial subsistence food for the northern Ojibwe Tribes, in particular for the Grand Portage Band of Lake Superior (Oliveira-Santos et al. 2021; Wolf et al. 2021; Severud et al. 2022, 2023*a*, *b*; Van de Vuurst et al. 2022). Additionally, many First Nations communities in Canada also rely on moose for subsistence, food security, and food sovereignty (Parlee et al. 2021). Beyond simply a source of food, moose hold cultural and spiritual importance to Indigenous communities as a traditional food and for ceremonial purposes like drum skins or hoof rattles (Tobias and Richmond 2014).

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Efforts to restore moose populations on the Grand Portage Reservation and ceded lands in what is now known as Minnesota have begun following >15 years of research to understand the factors affecting moose populations. Declining populations have been documented along the southern cusp of North American moose range in the northeastern and midwestern United States (Murray et al. 2006, Lenarz et al. 2010, Barber-Meyer and Mech 2016, Jones et al. 2019) and Ontario, Canada (Ranta and Lankester 2017). The mechanisms influencing these declines range from winter tick (*Dermacentor albipictus*) infestation and nutritional deficiency due to brainworm (*Parelaphostrongylus tenuis*) infection to wolf (*Canis lupus*) predation, with climate change an underlying exacerbating factor in most areas (Carstensen et al. 2018, Wolf et al. 2021, Severud et al. 2022, Van de Vuurst et al. 2022, Moore and Wolf et al. 2024).

As moose restoration efforts progress, co-stewardship with Indigenous nations is required to ensure subsistence lifeways and culture are maintained. This co-stewardship will allow for effective seventh generation planning, which is an Indigenous planning approach that incorporates a vision that we must leave the earth in a manner that our great grandchildren, seven generations from now may use it the same ways we do currently. Wildlife typically range over many jurisdictions and cooperative management by many entities is required for success (Ross et al. 2011). State-Tribal partnerships for moose management in northeastern Minnesota have been in place for many years, as moose range within the state is made up of state, county, and federal ownership, interspersed with private and Tribal lands. The majority of moose range is encompassed by ceded territory (Edwards et al. 2004). Minnesota and Tribal governments in the state cooperatively manage moose by jointly conducting a winter aerial survey and consultation on harvest (Edwards et al. 2004, Giudice 2023). Tribes and Tribal entities in Minnesota have also engaged in cooperative management well beyond population surveys and harvest consultations catalyze a better understanding of research and monitoring to explore population dynamics, underlying causes of mortality, and habitat-related work (Wolf et al. 2021; Oliveira-Santos et al. 2021; Van de Vuurst et al. 2022; Severud et al. 2022, 2023*b*; Moore and Wolf et al. 2024).

Co-stewardship is cooperative and co-equal, meaning collaborative engagement between Indigenous and non-Indigenous governmental land managers is required to meet shared interests in managing, conserving, and preserving natural and cultural resources. These practices currently occur under the primary responsibility of federal land managers, as Minnesota has historically been slow to recognize trust responsibility to Tribal nations; however, Minnesota is improving their engagement and recognition of co-management responsibilities in ceded areas with reserved usufructuary rights. Cooperative and collaborative engagements take a wide variety of forms based on the circumstances and applicable authorities. These may include sharing of technical expertise, combining Indigenous and non-Indigenous governmental capabilities to improve resource management and advance the responsibilities and interests of each, and making Tribal knowledge, experience, and perspectives integral to the public's experience of federal lands. Co-stewardship can be implemented through cooperative agreements, memoranda of understanding, self-governance agreements (including annual funding agreements), or less formalized mechanisms (Bureau of Indian Affairs 2022, Bureau of Land Management 2022).

In this paper we describe and discuss a study and synthesis on Indigenous co-stewardship of moose. We offer a synopsis and specific recommendations to advance restoration of moose in North America using core principles of a co-stewardship framework that we developed as part of the process. We describe the workshop and its products and develop a professional perspective of the author team composed of the workshop organizers.

METHODS

Background and summary of the North American Moose Co-stewardship project

On 25 May 2023, the Grand Portage Band held a workshop to advance international and Tribal co-stewardship of North American moose as a step toward forming a regional Tribal moose collaborative. This workshop was attended

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by participants of the 55th North American Moose Conference and Workshop held in Grand Portage, Minnesota. Grand Portage is an Indigenous community within Gichi-onigamiing (Grand Portage Indian Reservation), which is bordered by Lake Superior to the south and east, Ontario to the north, and state forest lands of Minnesota to the west.

The North American Moose Co-stewardship Workshop adapted the One Health Systems Mapping and Analysis Resource Toolkit (OH-SMART), an action planning tool to support collaborative analysis among diverse stakeholders (Vesterinen et al. 2019). The OH-SMART method has been used in the United States and in 29 countries internationally to strengthen how stakeholders coordinate in complex systems. We used OH-SMART to identify gaps in how moose populations are being managed. After we identified gaps, partners jointly identified what should happen to strengthen moose stewardship and made recommendations for improving co-stewardship moving forward. Thus, in using OH-SMART, the workshop objectives were to understand diverse values, perspectives, and needs for moose co-stewardship; understand which stakeholders are managing and researching moose, from community to cross-border organizations and agencies; understand diverse values, perspectives, and needs for moose co-stewardship; and understand how agencies and organizations are functioning in the system, who they are collaborating with, and how collaborations and programs are working (or not working). Following the workshop, we synthesized the results and drafted a summary that recommends core principles of Indigenous co-stewardship of moose; describes opportunities, challenges, and action steps; and identifies potential bottlenecks to success.

Workshop approach

Participants (111 in person, 33 remote) representing governmental agencies (local, state, Tribal, federal), private organizations, and academia from 8 Tribal nations and 6 other countries (USA [16 states], Canada [7 provinces], Italy, Japan, United Kingdom, Sweden) attended the planning workshop. The participants had wide-ranging expertise, perspectives, and experience in moose population and habitat stewardship, including population management, wildlife health, forestry, habitat, population dynamics, hunting perspectives, landscape genetics, and two-eyed seeing. Participant self-identified affiliations included 75 agency, 45 academic, 12 Tribal, 9 retired, 7 business, 7 non-governmental (NGO), 2 consultant, and 1 foundation.

Five keynote speakers opened the workshop to provide background and traditional cultural perspectives on efforts to establish international and Tribal co-stewardship of moose and to set the stage on opportunities and challenges in implementing a co-stewardship approach (Appendix, available in Supporting Information). Participants were assigned to working groups at 13 tables and 1 virtual breakout group, which were preassigned to be composed of different backgrounds and expertise in the moose management system. They were then asked to work together to complete 3 rounds of collaborative systems mapping and planning.

In round 1, working groups used a stakeholder mapping process to collaboratively identify diverse stakeholders at local, state or provincial, national, and international levels in the United States and Canada. They marked on the maps places where stakeholders had successful collaborations or best practices for co-stewardship of moose and points in the system where relationships have been difficult or confusing.

In round 2, working groups discussed 3 questions: What are the most pressing challenges in the co-stewardship of moose populations in the system represented at your table? What are some opportunities (existing programs and collaborations, funding, previous successes, etc.) to build on to advance co-stewardship of moose in your system? Of everything discussed, what are the 1–3 topics (challenges to address, opportunities to leverage) that you feel as a group are most important for moving forward and possible for participants to address? The groups' answers were recorded on flip chart paper and posted on walls around the room for further discussion.

In round 3, working groups developed action plans based on priority topics identified in round 2. They identified what should happen under each topic and what activities could be implemented to achieve those objectives.

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Stakeholder mapping and thematic analysis

We compiled stakeholder maps from round 1 into a single, composite, comprehensive stakeholder map (Appendix) using an online drawing app (draw.io, JGraph Ltd, Northampton, England). The stakeholder map summarized identified the interconnected relationships of stakeholders in the workshop and was used by the author team with long form notes to develop core principles of stewardship and for developing the report that identifies the thematic areas identified. We combined all repeating stakeholders among groups, recorded connections as an inclusive set among tables, and categorized them into 7 categories: international, Canada Tribal, United States Tribal, Canada national, United States national, Canada provincial, United States state, Canada local, United States local. Where tables had grouped stakeholders into categories, the composite map captured those categories using yellow rectangle boxes and grouped the related stakeholders together around those category boxes. If 2 tables used different terminology for similar groupings, we created a single, combined, or composite category term, and combined the groupings.

We transcribed all workshop outcomes, including presentations, notes, and flipcharts from the 3 rounds of collaborative decision-making to long form notes before further analysis. We formatted the results for each breakout group (tables and virtual) into an action plan in 3 categories: opportunities, challenges, and priority actions (described as solutions in the Appendix).

For opportunities, we then analyzed each category using deductive qualitative analysis to identify cross-cutting themes across all tables. In short, we compiled all opportunities into 1 comprehensive list, and grouped related opportunities into categories or themes and then specific areas of focus under each theme. We provided examples from tables that provided context for the theme for each area of focus where they were available to further elucidate the group's intent. We followed a similar approach to identify challenges and priority actions, and then provided examples (described fully in the Appendix).

RESULTS AND DISCUSSION

Core principles of Indigenous co-stewardship of moose

The first step in synthesizing the information collected from the co-stewardship workshop was to develop core principles of Indigenous moose co-stewardship that should be applied broadly to restore moose and can be applied in every environmental and natural resources concern for Indigenous people. The principles were developed by the author-team using the stakeholder map and the tables of challenges, opportunities, and priority actions summarized from long form notes. These principles reflect the highest priority of supporting Indigenous lifeways, the incorporation of Indigenous perspectives into policy, and the commitment to meeting federal trust responsibilities to Indigenous nations.

1. Recognition of Indigenous sovereignty. Co-stewardship must begin with a recognition of the sovereignty of Indigenous nations and their inherent rights to manage, conserve, and preserve natural and cultural resources within their ancestral lands. This recognition is based on historical treaties, international agreements, and the principle that Indigenous Peoples have a unique and vital role in sustainable land management. Supporting the lifeways of Indigenous Peoples and their environment must be the highest priority.
2. Shared responsibility. Co-stewardship is a shared responsibility between Indigenous nations, federal governments, and state governments. All parties must work collaboratively and cooperatively to manage and protect moose populations and their habitats, recognizing that these resources are essential for both cultural and ecological reasons. Federal trust responsibilities to Indigenous nations must be met.

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3. Cultural and ecological respect. Co-stewardship should honor the cultural significance of moose to Indigenous nations and recognize the ecological importance of moose within the broader ecosystem. It should involve a deep respect for traditional knowledge, practices, and values, and scientific insights and conservation principles. This respect should be incorporated into the broadest policy levels.
4. Inclusive decision-making. Co-stewardship requires inclusive and equitable decision-making processes that involve meaningful consultation and consent from Indigenous nations. This includes the active involvement of Indigenous elected leaders, elders, and community members in all aspects of moose management, from setting harvest quotas to developing conservation strategies.
5. Resource sharing and capacity building. Co-stewardship should involve the sharing of resources and knowledge between Indigenous and non-Indigenous partners. This may include the transfer of technical expertise, financial resources, and collaborative capacity-building efforts to empower Indigenous nations to take a leading role in moose conservation and management.
6. Adaptive management and sustainability. Co-stewardship should embrace adaptive management principles, where management strategies are continuously evaluated, adjusted, and improved based on new information and changing conditions. The goal is to ensure the long-term sustainability of moose populations, recognizing that their health is intertwined with the health of the ecosystems they inhabit.

Opportunities

We strategically outlined the unmet potential for enhancing co-stewardship in North American moose management (Table 1). Six broad themes of opportunities emerged including 1) new relationships and partnerships that can be developed, 2) new and expanding constituencies and perceptions that can be constructively used, 3) development and expansion of mutually beneficial opportunities in managing habitat and moose, 4) identifying communication and collaborative models that currently exist and using those approaches more broadly, 5) resources are currently available and can be more effectively used, and 6) co-stewardship can leverage those existing resources for more impact.

New relationships and partnerships

New relationships and partnerships included the specific areas of new Tribal-university research and Tribal-state relationships. For example, Joint Secretarial Order 3403 (Fulfilling the Trust Responsibility to Indian Tribes in the Stewardship of Federal Lands and Waters) emphasizes the fostering of inclusive co-stewardship (U.S. Department of the Interior and U.S. Department of Agriculture 2021). Participants also recognized that expansion of thoughtful and effective collaborations can improve collaborative management. An example of such collaborations is the development of the America the Beautiful moose habitat project focused on developing large tracts of land, 4,000–20,000 ha for creation of moose habitat in Minnesota (Dawson et al. 2022). This project includes Tribal moose experts, state moose biologists, federal, Tribal, and state foresters, and social scientists to host workshops and synthesize information gathered to identify areas of relative consensus.

New and expanding constituencies

Some of the specific areas mentioned to expand constituencies were a growing public awareness of climate change, treaty obligations of federal and state governments, and conservation in general. The growing interest in and awareness of these issues can be used to increase co-stewardship of wildlife. For example, awareness of effects of

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TABLE 1 Workshop participant-identified solutions and recommendations to opportunities, challenges, and priority actions related to North American moose co-stewardship. Information was gathered from participants during the 55th North American Moose Conference and Workshop, Grand Portage, Minnesota, 25 May 2023.

Category	Solutions and recommendations
Opportunities	Develop new relationships and partnerships (e.g., Tribal-university research, Tribal-state relationships) Expand of constituencies and leveraging public awareness (e.g., climate change, treaty obligations) Development and expansion of mutually beneficial opportunities Use existing communication and collaboration models (e.g., Minnesota Moose Habitat Collaborative) Resources are available Co-stewardship can leverage existing resources for more impact
Challenges	Resource and support limitations for co-stewardship Tensions among stakeholders (consumptive vs. non-consumptive, differing cultural perceptions) Unaddressed research and management needs Barriers to collaboration (communication, trust, bureaucratic hurdles) Mosaic of land ownership and regulation Lack of consensus on best practices
Priority actions	Building trusting relationships Science-based and shared moose population management Enhancing communication, coordination, and action and formalizing co-stewardship frameworks Developing regional landscape-scale strategies Ensuring inclusive and equitable engagement

climate change through products from Intergovernmental Panel on Climate Change (2023) describe impacts to biodiversity and ecosystem services (Calvin et al. 2023). The 5th National Climate Assessment specifically describes effects on moose populations and on Indigenous cultures like the Ojibwe of Minnesota (McElwee et al. 2023). Increased recognition of treaty obligations is evidenced in guidance such as Secretarial Order 3403 (U.S. Department of the Interior and U.S. Department of Agriculture 2021), which specifically discusses trust obligations and co-stewardship with Indigenous nations. The increased public awareness of these issues will allow for improved public education campaigns and greater gains in improving the environment and our natural resources.

Development and expansion of mutually beneficial opportunities

Participants discussed expanding mutually beneficial scenarios including managing habitats and moose and growing research collaborations. In many instances, timber management can benefit moose (Johnson and Rea 2023) and other boreal species (Thompson et al. 2003). As an iconic species, moose can mobilize support from users of natural resources. State-Tribal partnerships can create these mutually beneficial scenarios. Additionally, there are increasing numbers of university-Tribal partnerships and collaborations around moose research and other topics. Grand Portage Band of Lake Superior Chippewa's Division of Biology and Environment has partnered with the University of Minnesota's College of Veterinary Medicine since 2015, and subsequently, State University of New York Environmental Science and Forestry, University of Minnesota's Natural Resources Research Institute, Michigan State University, Iowa State University, and South Dakota State University. In each of these research relationships, a data sharing and ownership agreement is in place that protects Tribal data sovereignty (Moore et al. 2015), underscoring the commitment of all parties to formalizing relationships. Successful implementation of such agreements can also increase the level of trust among parties, particularly Tribes, who have been at a disadvantage in the past without such agreements in place.

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Existing communication and collaboratives

Another theme was that examples of effective communication and collaboratives exist and can be mirrored. For example, the United States Forest Service and The Nature Conservancy work together on forest management projects (www.nature.org/en-us/newsroom/usfs-tnc-five-year-cooperative-agreement/, accessed 16 May 2024). The Minnesota Moose Habitat Collaborative is another example (Minnesota Department of Natural Resources [MNDNR] 2016). State and federal partners work together in forestry projects often with input from Tribes that have ceded territory rights on those public lands. Experts in moose ecology meet annually to share the state of knowledge of moose management at the North American Moose Conference and Workshop, and every 5 years at the International Moose Symposium. These meetings allow the sharing of knowledge and ideas. Outcomes of the meetings are often shared publicly in peer-reviewed journals (Boer 1992, Moen et al. 2000, Editors 2020).

Intertribal organizations, such as the 1854 Treaty Authority and Great Lakes Indian Fish and Wildlife Commission also have co-management responsibilities over ceded public lands and help advance Tribal goals to improve natural resources. For example, the 1854 Treaty Authority and Fond du Lac Band of Lake Superior Chippewa have worked together with the Minnesota Department of Natural Resources to conduct an aerial survey of moose from 1996 to present (Edwards et al. 2004, DelGiudice and Giudice 2022).

Resources are available and can be more effectively used

Another major theme to emerge during the workshop was that underused resources are available for co-stewardship. Specifically in Minnesota, the Legislative-Citizen Commission on Minnesota's Resources and the Lessard-Sams Outdoor Heritage Council are potential state funding opportunities for moose research and habitat restoration. There are also trusts and foundations willing to fund moose management. Federal funding sources for co-stewardship and co-management are available through sources such as the Great Lakes Fish and Wildlife Restoration Act, Great Lakes Restoration Initiative, America the Beautiful Challenge grants, Bureau of Indian Affairs Climate Resilience grants, and National Science Foundation programs. There are also federal funds specifically dedicated to Tribal nations such as the Distinct Tribal Program under the Great Lakes Restoration Initiative.

Co-stewardship can improve existing resources for more impact

The working groups envisioned that co-stewardship can leverage existing resources for more impact. Cutting through bureaucratic red tape and accessing each other's resources can increase efficiency and effect. Also, because moose are wide-ranging animals that often traverse multiple jurisdictions, co-stewardship allows for opportunities for large-scale management, which is required for moose. Landscape-level prescribed fire or timber management could be applied and help curtail some maladies of moose while enhancing food and cover needs; for example, application of prescribed fire may be effective in controlling populations of winter ticks on moose, while simultaneously creating young forest and early successional growth required for moose foraging. Similarly, logging practices that restore young forest can also be effective in controlling gastropods that are vectors for brainworm in moose (Severud et al. 2023*b*)

Challenges

The next section of our work identified challenges impeding the efficient management and conservation of moose (Table 1). We analyzed each barrier and offered potential solutions (Table 2; Appendix), resulting in a

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TABLE 2 Solutions to challenges to Indigenous co-stewardship identified by participants of the 55th North American Moose Conference and Workshop, Grand Portage, Minnesota, 25 May 2023.

Barrier to Indigenous co-stewardship of moose	Solutions
Lack of resources and support for co-stewardship	Resource pooling among Tribal nations, state, federal, and academic institutions Grant and funding opportunities for collaborative initiatives Creative habitat development approaches supporting the logging industry Capacity building for Tribal staff in moose management
Tension, trade-offs, and differing perceptions	Stakeholder engagement to promote dialogue and understanding Multi-species management considering the interdependencies among species Public awareness campaigns to bridge gaps between perspectives
Unaddressed research and management needs	Identify and prioritize unaddressed research needs Dedicate funding and staff for research and management Climate adaptation research to address climate change impacts Health and habitat management for ecosystem health Cross-border collaboration for large-scale habitat management
Barriers to collaboration	Streamlined administrative processes to enhance efficiency Enhanced trust-building through conflict resolution and awareness campaigns Cross-cultural competency and understanding among agency employees Policy reforms and Tribal engagement for inclusive decision-making Applied research and altruistic projects focusing on practical solutions Conflict resolution between hunter groups
Academic and Indigenous barriers	Establish clear guidelines and agreements Fund Indigenous-led research initiatives Foster genuine involvement of Tribal governments Prioritize applied research for practical outcomes Promote collaborative governancePromote ethical research practices Recognize non-publishable outcomes Establish accountability mechanisms
Non-governmental organization (NGO) and Indigenous barriers	Education of NGO leadership about Tribes and sovereignty integration of mission areas to include Tribal perspectives Co-development of collaboration areas for mutual interest Establishment of moose constituency group Mandatory Indigenous engagement criteria for NGOs Indigenous advisory boards for ongoing guidance
Mosaic of land ownership, management, and regulation	Integrated land management plans across land boundaries Coordination and consistency in land management goals
Lack of consensus on co-stewardship best practices for landscape management	Best practices guidelines for co-stewardship Interagency workshops to disseminate best practices

comprehensive and prescriptive understanding of the issues hindering effective co-stewardship. Six challenges were identified by participants: 1) lack of resources and support for co-stewardship; 2) tension, trade-offs, and differing perceptions among other constituents and stakeholders; 3) there are many research and management needs; 4) barriers exist to collaboration among stakeholder groups particularly for Tribal partners; 5) there is a

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mosaic of land ownership, management, and regulation; and 6) there is lack of consensus on co-stewardship best practices. Solutions were proposed to bridge the gaps in understanding, resources, and collaboration that currently impede Indigenous co-stewardship of moose (Appendix), emphasizing the need for a holistic, inclusive, and adaptive approach to wildlife conservation (Table 2).

Lack of resources and support for co-stewardship

A specific challenging area identified was lack of resources for habitat management. This was cited as due to a lack of funding, the limited number of contractors who would take on specialty forestry operations (e.g., selective cutting), distance to markets and mills, and lessening demand for paper and pulp products. A potential opportunity would be to hire contractors that can simply develop moose habitat, where contractors are paid to cut spruce budworm (*Choristoneura fumiferana*)-damaged forests, which will then become moose habitat. Some groups also identified a lack of resources for moose management, although this conflicts with some groups' opinions listed in the opportunities. These contradictions may highlight different jurisdictions where moose are prioritized as a game or conservation species, and where conservation dollars are not available for moose or any other subsistence species. Another complication is that resources are partitioned among stakeholders and thus are not consolidated.

Tribal staffing limitations are a recurring issue in Indian Country broadly and the numbers of dedicated Tribal staff working on moose in Minnesota is an example. There are 3 Tribal biologists that work on moose serving an area nearly 2,000,000 ha in size, and the moose-focused work is only a fraction of the overall fisheries and wildlife responsibilities of those positions. These biologists work actively on all terrestrial and aquatic subsistence species.

Tension, trade-offs, and differing perceptions

The tensions among various stakeholders in moose landscapes arise from several factors. A lack of trust by the public of government and academic work is compounded when that work is not implemented by managers. Resource users were also cited as often being in conflict, which puts agencies serving those stakeholders also in conflict. Another example is the tension and conflict that can arise between extraction industries and Tribal nations. Minnesota Forest Industries pressures Minnesota legislators to manage forests mostly for timber volume rather than multiple uses and habitat. This sometimes causes conflict between the MNDNR Forestry and Tribal natural resource agencies, and within MNDNR Forestry and Wildlife Divisions. There are also tensions among various single-species advocacy groups. White-tailed deer (*Odocoileus virginianus*), moose, and wolves all have ardent supporters, and perceptions that all 3 species can coexist at levels desired by various stakeholder groups in the same area without active management can be problematic (e.g., wolf predation on moose calves, parasite transmission from deer to moose).

Unaddressed research and management needs

There are also research and management needs that have not been addressed. Climate change affects moose in myriad ways, and working groups indicated that there needs to be a better understanding of how to manage climate impacts to the ecosystem. Moose health is often a factor in populations declining or stagnating (Murray et al. 2006, DelGiudice et al. 2011). Research around parasite management would be useful to determine effective approaches. A better understanding of management strategies for moose habitat in the face of climate change is also needed. There was also a recognized need to understand the relationships between insect outbreaks, forest fires, and timber management, and how these can be leveraged to restore moose habitat. In addition, there needs to be a better

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understanding, collaboration, and coordination for large-scale habitat management across borders. A recently developed Ontario–Manitoba–Minnesota moose collaborative is an example of a group interested in consistent and international research and management projects dedicated to moose.

Barriers exist to collaboration

Barriers to collaboration were identified by participants, chief among them trust and communication. Several key challenges were identified in the engagement between Indigenous nations and the state of Minnesota, specifically around natural resources management, including moose management. These challenges stem from a historical context of mistrust and communication barriers. The distrust results from a long history of Tribal-state relationships characterized by conflict, misunderstandings, and the states' refusal to acknowledge the reserved usufructuary rights and management authority of Indigenous nations as stipulated in treaties. This refusal has directly influenced the collaboration efforts on environmental conservation and resource management initiatives.

The complexities of these challenges are further exacerbated by the legal and institutional frameworks within which these engagements occur. Despite landmark federal cases in the Great Lakes region, such as United States v. Michigan (1979, 1985) and Minnesota v. Mille Lacs Band of Chippewa Indians (1999), which have affirmed the sovereignty and treaty rights of Indigenous nations, translating these judicial recognitions into consistent and respectful state-level policy and action has been fraught with difficulties. These cases underscored the enduring legitimacy of federal-Tribal treaties and reserved rights, setting a legal precedent for the recognition and enforcement of Indigenous treaty rights. The application of these rulings in practice has seen intermittent progress at best, with states often slow to align their policies and actions with the federal judiciary's direction.

This inconsistency can be attributed to several factors, including a lack of awareness or understanding of Indigenous treaty rights among state officials, institutional inertia, and resistance to change. The challenges are not merely legal but also involve deeply ingrained perceptions and attitudes that can hinder meaningful dialogue and cooperation. The path to overcoming these barriers requires an approach that goes beyond court rulings and necessitates building genuine relationships based on mutual respect and understanding, enhancing communication, and fostering trust. This involves acknowledging the historical injustices and working collaboratively with Indigenous nations to co-manage natural resources in a way that respects their sovereignty, cultural practices, and economic needs.

Moreover, addressing these challenges requires a concerted effort to educate all stakeholders about the historical and legal contexts of Indigenous treaties and rights. There is a need for institutional change within state agencies to integrate Indigenous perspectives and knowledge into natural resources management. Such changes could facilitate more inclusive and effective co-management frameworks that recognize and uphold the treaty rights of Indigenous nations, ultimately leading to more sustainable and equitable outcomes for Indigenous communities and the broader society. While progress in Minnesota has been slow, the current executive branch has selected an Indigenous Lieutenant Governor and relationships have improved rapidly.

Tensions among state, federal, and Tribal governments and agencies was another broad theme of barriers to Indigenous co-stewardship. Many state and federal agency employees described a lack of awareness of Tribal sovereignty and rights, or state and provincial governments making moose management decisions without regarding Tribal or Indigenous viewpoints. A general lack of intercultural understanding and competency was also identified as a barrier, though it was acknowledged that this could be broadened to include different cultures, policies, and political systems among nations, states, provinces, and Tribal nations. A specific example was the various protections afforded to wolves in different jurisdictions and the cultural reverence or revilement of wolves in those jurisdictions.

Participants additionally identified there is a lack of coordination and communication within agencies, which in turn limits inter-agency coordination and communication. Some agencies have bureaucratic systems that bog down and create barriers to collaboration.

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A similar history of difficult Tribal-academic relationships and an overall lack of trust has led to limited academic engagement with Tribes on moose research. Academic institutions, in their purported responsibility to Indigenous nations, find themselves ensnared in a multifaceted conflict of interest. This conflict is emblematic of a deep-rooted structural and ideological issue, where administrative and philosophical paradigms within these institutions obstruct genuine progress required for developing moose research, restoration, and management practices in collaboration with Tribal nations. For example, many Indigenous nations have distrust of academia related to the boarding school era initially, further exacerbated by exploitative research practices in more modern times as evidenced in the TRUTH project (TRUTH Project 2023). Additionally, many state-run academic institutions prioritize theoretical or basic research, rendering collaboration impractical for Tribal nations whose needs demand tangible outcomes and practical applications.

Administrative burdens and lack of funding exacerbate challenges for Indigenous nations to collaborate with academic institutions. Many United States universities are Federal Land Grant institutions and take great pains to acknowledge their burden of responsibility to the Tribes on whose land they sit, but the extent of their acknowledgment seems to rest at the formation of statements of acknowledgment rather than actually leading or making positive change. Federal Land Grant institutions have trust responsibilities to Indigenous Peoples codified by law and upheld by the Supreme Court (TRUTH Project 2023). In collaborations between academia and Tribal nations, there is an expectation that Tribes contract academics and their institutions for research work despite the claims of acknowledgment and burdens of responsibility virtuously expressed in those statements. A specific example identified university indirect cost overhead as a barrier to Tribal-university partnerships. It was suggested that there is a need to develop administratively streamlined and cost-effective mechanisms for collaborative projects. The conflict between academia and Indigenous nations becomes starkly evident when Tribal nations seek to partner with academia and discover that they are unfairly burdened with the costs of research, including the time and efforts of academic faculty and their students, and the indirect expenses of the institutions themselves. There are examples of institutions waiving indirect cost when research funds move from Tribal organizations to universities (e.g., South Dakota State University, Michigan State University). Disillusionment can set in as the resulting products prove to be primarily theoretical or basic scientific findings, devoid of any real-world applications. Academic faculty may be naively indoctrinated by this conflicted paradigm and fail to recognize that conflict exists at all. The academic system's reward structure with peer-reviewed research and the pursuit of publishing accolades further exacerbates the problem, discouraging altruistic projects that could genuinely benefit Tribes but may not yield publishable units.

In a concerning but disappointing theme, Indigenous nations face significant hurdles when attempting to engage with non-governmental organizations (NGOs), foundations, and charitable trusts for moose restoration. These challenges are not merely unfortunate; they point to a pervasive conflict of interest that perpetuates the marginalization of Indigenous voices. Furthermore, there are occasional direct conflicts of interest with NGOs and Indigenous co-stewardship principles. An example is a Minnesota deer hunting association participating in a broad-scale moose habitat collaborative; there is a directly conflict because they are primarily a deer organization and actively advocate for greater deer numbers in northern regions of core moose range despite significant risks of transmission of brainworm from deer to moose.

The impediments mirror the issues related to the lack of understanding regarding sovereignty, treaty rights, and the inherent sovereignty of Indigenous nations; however, some are specific to those types of entities. Most NGOs, trusts, and foundations have very specific missions and those are nearly always defined absent Indigenous perspective or involvement except for a very few Indigenous-led organizations. In their pursuit of objectives, these organizations often employ misaligned agendas as a pretext for engaging with Tribes, with Tribes frequently being instrumentalized to lend an air of legitimacy to the matters at hand. Many NGOs will ask Indigenous nations to endorse amicus briefs or participate in lawsuits, thereby implying the involvement of a sovereign governmental entity in the litigation. Another common issue occurs when individual Tribal members may be recruited to present their views and those perspectives are often taken or put forth as the Tribal perspective or a Tribe's position.

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This tokenism is problematic when that individual may have no authority to speak on behalf of a Tribe and also ignores that opinion amongst individual Tribal members can run the full gamut of perspectives on any given issue. While such collaborations can indeed lead to productive outcomes and environmental protections, a glaring issue is that the agendas of these NGOs seldom encompass or incorporate Tribal viewpoints, almost never seek Tribal input in shaping their missions, and only use Tribes when it serves their convenience in a one-sided fashion. Financing collaborations between Tribes and NGOs, foundations, and trusts is another challenging area because rarely are funds dedicated to Tribal work or natural resources. Most foundations and trust agendas are not specifically aligned with Tribal subsistence needs.

Mosaic of land ownership, management, and regulation

The mosaic of land ownership, management, and regulation was also noted as a challenge for a wide-ranging species such as moose. Regional strategic planning was considered difficult when the landscape is owned by private citizens, industry, nonprofits and NGOs, or federal, state, county, or Tribal agencies. The different regulations, policies, and limitations on use among the different landowners create conflict and misunderstandings, making coherent large-scale habitat co-stewardship strategy challenging. The missions of each agency typically differ and have different goals for the purpose of the land, and in some cases, may not benefit moose.

Lack of consensus on co-stewardship best practices for landscape management

The lack of consensus on co-stewardship best practices as identified by participants may actually be a lack of knowledge of best practices. Additionally, there is a lack of consensus on what is good moose habitat. Each agency manages forests by their own objectives, mandates, and goals, which may not be moose-related. Some agencies manage for harvestable timber, some may just want more moose, and others may want healthy moose. Addressing these discrepancies to achieve broad consensus that is aligned with moose restoration is important.

Priority actions

The priority actions are actionable steps, acknowledging the importance of not only scientific advancements but also the need for robust partnerships, communication, and inclusivity in ensuring effective co-stewardship (Table 1). We emphasized collaborative decision-making, transparent actions, and the leveraging of technological advancements to enhance moose population management. Five broad areas were identified for priority improvements: 1) building trusting relationships; 2) science-based and shared moose population co-stewardship; 3) improving communication, coordination, and action for co-stewardship broadly and then formalizing co-stewardship arrangements; 4) developing regional landscape-scale co-stewardship strategies; and 5) inclusive and equitable engagement of all constituencies and stakeholders. Participants identified priority issues, what should be done, and actions to be achieved.

Developing trusting relationships

This synthesis emphasizes the importance of trust and collaboration between Indigenous nations and non-Indigenous entities (governments, institutions, organizations) in stewarding and conserving moose populations in North America. Indigenous Peoples have a long history of sustainable environmental stewardship, deeply rooted in

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their cultural and spiritual values, which has been overlooked or undermined because of colonization, legal frameworks, and policies that did not recognize their sovereign rights or traditional knowledge. Recognition of Indigenous sovereignty, rights, and traditional knowledge is central to the stewardship and conservation of moose. Resolving trust issues requires a shift towards genuine collaboration, shared governance, and an integrative approach that respects Indigenous knowledge and scientific research.

Tribal-state relationships are strained because of a history of exclusion and colonialism. Additionally, states were not signatory to the treaties with Tribes that occurred prior to statehood in most cases and thus have not assumed or accepted a trust responsibility to Tribes in the same manner as the federal government. Additionally, the United States federal government has historically taken the perspective that states retain general ownership of, and management authority for, fish and wildlife resources within their boundaries, only stepping in during situations where a specific species may be threatened. There is a need for the United States Government to publicly, and often, make the point that Tribes have never relinquished their role as managers of resources and ensure that states and others accept that fact. To ensure effective decision-making and policy implementation, it is important to involve Tribes in state processes, ensuring them a seat at the decision-making table and ensuring meaningful consultations. This should be facilitated through government-to-government consultations and continuous and cooperative partnerships between Tribes and state or provincial agencies. Such partnerships should ideally involve shared funding for collaborative projects and receive support from leadership on both sides, including Tribes and state governments, to foster a spirit of collaboration and consensus. To build the capacity of state agency staff in fostering respectful partnerships, training should be provided. Collaborative monitoring and data dissemination, as seen in examples like Nova Scotia's Tribal and non-Tribal moose check stations, should be prioritized to ensure trust in data and decisions (uinr.ca/programs/moose/, accessed 16 May 2024). Seeking shared funding opportunities, such as co-principal investigators on grants or co-management of funds, is another way to strengthen partnerships between Tribes and state agencies. The creation of collaborative working groups and systems, like the Indigenous Stewardship forums in British Columbia, Canada, can foster ongoing partnership. Regular communication and collaborative research, exemplified by citizen science, should also be encouraged. Overall, these strategies were identified as tangible approaches to promoting meaningful and effective collaboration between Tribes and state or provincial agencies.

A priority issue identified was to implement strategies to build trust and community among co-stewardship stakeholders and rights-holders. Participants stated it is essential to follow through on commitments made by all agencies involved. Transparency in actions and decisions among co-stewardship partners should be rigorously upheld, with a particular focus on data sharing, co-development, and co-authorship of project outputs. Collaborative knowledge should be fostered, facilitating the circular integration of insights and expertise among researchers, community members, implementers, and decision-makers. A data-sharing platform could facilitate interaction among agencies, hunters, Tribal staff, and other stakeholders, allowing them to report locations, use tools like eMoose and iNaturalist, and share essential data. Integrated discussions should be proactively encouraged throughout the various project phases, starting with project planning. This approach ensures that multiple perspectives are included in conversations. To support partnership building, it is recommended to provide travel funds and translation support to staff, enabling effective collaboration and knowledge exchange.

Science-based and shared moose population co-stewardship

Several areas of research were identified by participants to advance moose population co-stewardship. Following many studies finding parasites are important factors affecting moose populations (Carstensen et al. 2018, Grunenwald et al. 2018, Hoy et al. 2021, Oliveira-Santos et al. 2021), a call to strengthen parasite management was made by participants. This included a need for more funding for research and application of moose-parasite research.

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As this workshop was conducted in Minnesota, many participants were local and called for the establishment of a moose co-stewardship collaborative model in Minnesota. For example, the Minnesota Moose Management Plan was out of date at the time of the meeting (MNDNR 2012). An updated, collaboratively developed moose plan was recommended, which envisions recovery through a shared adaptive framework.

White papers that result from the annual North American Moose Conference and Workshop and the International Moose Symposium would be helpful for applied management recommendations. These meetings convene local moose managers and help them understand effective moose management.

Robust partnerships among academic, Tribal, state, academic, and other implementing partners need to be fostered and strongly supported by executives and administrative actions to best enable effective staff-level collaborations. The research being undertaken should be timely and meaningful so it can assist with decision making and natural resources management action. The findings need to be translatable and digestible to inform decision makers at all agencies. Through the use of data sharing and ownership agreements (Moore et al. 2015), academic institutions have an obligation to share information with partners and collaborators. In addition, communications departments should conduct outreach to Tribes and other agencies involved in moose management.

Technological advances can better inform habitat management, and these methods should be fostered and used in decision making. Additionally, these methods can help to understand the effect of management actions on wildlife populations. Technological advancements may include tools like satellite global positioning system (GPS) collars for better determination of vital statistics and habitat use of wildlife. For example, using GPS-collar technology measuring moose responses to prescribed fire or wildfire can be evaluated and then considered in management plans (Mumma et al. 2024). New analytical tools like deep learning and artificial intelligence will be helpful for data analysis and strengthen predictive abilities.

Improving communication, coordination, and action for co-stewardship

The initial approach identified was to establish a shared understanding and goal or set of objectives across diverse stakeholders and rights-holders. Short- and long-term goals need to be collectively identified around common holistic goals. To achieve this, a shared understanding of the state of knowledge around moose management needs to be communicated to all parties. Any set of objectives will need to be prioritized, and participants should be clear that some objectives may conflict. A conflict negotiator may therefore be needed.

To ensure the sustained success of collaborative efforts, it is essential to establish continuous communication, partnerships, and collaborations among stakeholders. These systems should facilitate interaction and ensure that leadership and decision-makers maintain ongoing communication. It is also important to set up mechanisms for enduring cross-border collaborations that extend beyond one-time meetings. Regular meetings should be established among the leaders of key stakeholder organizations to promote cohesion and shared goals. To maximize cooperation, leaders of adjacent lands within the area of interest should be brought together to create a collaborative timetable for planning milestones. This will allow for coordinated efforts and efficient resource allocation. The formation of cross-border and cross-agency working groups, involving professionals like foresters, wildlife managers, and Tribal communities, should be prioritized.

A formal co-stewardship framework must be established for management ideas to be put into action. A shared framework for collaboration and coordination will help to mobilize funds. A dedicated lobbyist could advocate for moose-related legislation and funding. This framework should aim to minimize structural constraints on collaboration and coordination to be effective. Intercultural competencies, awareness, and skills will be needed for the participants. Formal agreements can help justify time and resources spent on co-stewardship activities.

The dissemination of the successes and needs of any collaborative co-stewardship framework will help promote participation. Outreach will also highlight the importance of the collaboration to constituents, policy makers, and partners.

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Developing regional landscape-scale co-stewardship strategies

Three priority landscape-scale issues emerged from the workshop participants: 1) foster a collaborative landscape-scale model for co-stewardship (e.g., across federal, Tribal, provincial or state, non-profit, private landowners, industry); 2) support all stakeholders to achieve shared co-stewardship goals; and 3) understand and effectively leverage the mosaic of land ownership, land management, and associated regulations and restrictions that makes up landscape-scale moose habitat. To foster a collaborative landscape-scale model, shared objectives must be developed with accountability and follow-through by participants. This will also require development of systems for supporting landscape-scale collaboration, decision making, and funding mobilization. There are several examples in Minnesota that can be built upon, including the Swamp River Wildlife Management Area, which is currently co-managed by the United States Forest Service and the 1854 Treaty Authority for conservation of waterfowl and moose. Workshop participants suggested holding a planning session with >50 diverse moose stakeholders and rights-holders within Minnesota moose range and identified an America the Beautiful Challenge grant that could fund this meeting. To support all stakeholders and rights-holders to achieve shared co-stewardship goals, there needs to be outreach and engagement of the public and vested stakeholders, including landowners in moose range. Private citizen education campaigns for how to manage their lands for moose, and incentives to do so, could be tailored to different groups.

The mosaic of land ownership in moose range can be a barrier but also an opportunity. To understand and effectively leverage this mosaic, organizations can use their different regulatory environments to optimize moose management, land use, and community engagement.

Throughout this process, there must be an assurance of a diverse, equitable, and inclusive workspace and environment across all key stakeholders and rights-holders. The representation of Tribal students and other underrepresented communities in academia and agencies needs to be addressed. Hiring processes at all levels should be inclusive and non-biased. Recruitment, scholarships, more collaborations between Tribal Colleges and Universities and large universities should be prioritized.

Bottlenecks and final recommendations

Our highest priority for the implementation phase of co-stewardship of moose is a focus on formally designating core moose range(s) as priority areas for legislative or executive policy action. We must ensure that where moose populations are in decline, they are recognized as a priority species for their intrinsic value, as an indicator of a functional boreal ecosystem, and as a vital subsistence species to Indigenous nations.

There are significant challenges and opportunities to restore moose with a framework fully dependent on principles of co-stewardship. Optimization on either side of a bottleneck does nothing to improve overall efficiency and thus each limitation must be addressed in priority of the effect on restoration. We identified 8 factors most important to advancing restoration of moose in a co-stewardship framework.

1. Identify moose range and prioritize legislation. We recommend a focus on formally designating core moose ranges as priority areas for legislative or executive policy action. Where moose populations are in decline and in areas where they are used for subsistence, they must be recognized as a priority species. This prioritization includes advocating for policies that consider the unique habitat, health, and fitness needs of moose, particularly in regions where timber, predator, and deer management practices are in conflict with moose conservation (MNDNR 2018).

   If moose are to be restored, the species must be identified as the highest natural resources priority in core parts of their range. Currently, natural resources management priorities impede moose restoration at best and are destructive at worst. Forest management priorities currently encourage harvestable timber, which results in low-quality forage for moose, as they require both young forest and shrub and browse species. Management for harvestable deer leads to disease risk by brainworm to moose and further supports higher wolf densities than in the past (Barber-Meyer and

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Mech 2016). Protections on wolf harvest by the Endangered Species Act in the United States precludes wolf harvest and reductions in predation-induced mortality on moose calves, and limited harvest on black bears (*Ursus americanus*) exacerbates the additional predation mortality on calves that ensures nearly no recruitment of juvenile moose (Moore et al. 2024).

2. Allocate resources for active restoration. Tribal, federal, and state governments and stakeholders should dedicate significant financial and staffing resources to actively restore moose populations by implementing known conservation strategies and management practices.
3. Holistic restoration approach. Stakeholders should adopt a holistic approach that acknowledges the complexity of moose restoration. We recommend restoration strategies that consider the interactions between land management practices, species affecting moose populations (such as deer, wolves, bears, and gastropods [as parasite intermediate hosts]), and the broader ecosystem.
4. Formalize a co-stewardship framework. A formal co-stewardship framework that aligns various management ideas into actionable strategies is necessary. This framework should encourage collaboration among diverse stakeholders to achieve shared goals. Representative Tribal membership is necessary.
5. Incorporate Indigenous perspective and science-based approach. Indigenous perspectives in conjunction with a Western science-based approach should be used for managing habitats and associated boreal species. Stakeholders should identify optimal ecosystem management approaches, particularly focused on moose-centric ecosystems in core moose areas, integrating Indigenous knowledge with scientific findings.
6. Support applied science mechanisms. We recommend that resources are provided to foster applied science in collaboration with academic partners. This will ensure dedicated time and intellectual effort toward the specific goal of moose restoration, facilitating the implementation of scientific findings in conservation practices.
7. Large-scale restoration. Moose management efforts should consider the largest feasible spatial scale for moose restoration to achieve a comprehensive and extensive effect on population recovery.
8. Support harvest management. Moose management plans should encourage and support Indigenous subsistence harvest of moose, emphasizing sustainable practices. Similarly, they should promote non-Indigenous harvest while ensuring it is managed to maintain a healthy moose population.

These final recommendations align with the priority actions discussed in the synthesis, emphasizing the need for policy action, resource dedication, holistic approaches, and the incorporation of Indigenous perspectives and scientific methodologies to ensure a successful moose restoration program.

RESEARCH IMPLICATIONS

The findings and recommendations in this commentary have significant implications for the Indigenization of ecological research and wildlife co-stewardship. Our work underscores the necessity of integrating Indigenous knowledge systems with Western scientific approaches to create more effective and sustainable wildlife management practices. This integration can lead to more holistic and inclusive strategies that honor cultural values and ensure ecological integrity.

The core principles of Indigenous co-stewardship developed in this study provide a framework that can be adapted for various species and ecosystems far beyond moose. Implementing these principles will improve management outcomes by fostering collaborative, inclusive decision-making processes that respect and incorporate Indigenous sovereignty, cultural values, and ecological insights.

The establishment of formal co-stewardship frameworks can serve as a model for other regions and countries, promoting more effective and just conservation practices globally. Our recommendations highlight the need for policy reforms that recognize and enforce Indigenous treaty rights and promote and enhance shared governance leading to equitable management of natural resources and ensuring that Indigenous nations have a meaningful role in managing their ancestral lands and resources.

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The emphasis on adaptive management and the need for continuous evaluation and improvement based on new information aligns with principles of modern ecological research. This approach can enhance the robustness of wildlife research by incorporating long-term monitoring and flexibility to adapt to changing environmental conditions. Additionally, fostering collaborations between Indigenous and non-Indigenous researchers will lead to innovative research methodologies and a more comprehensive understanding of ecological systems.

The focus on resource sharing and capacity building can lead to strengthened research and management capabilities for Indigenous communities. By providing technical expertise, financial resources, and collaborative opportunities, researchers and policymakers can support the empowerment of Indigenous nations to take a leading role in wildlife conservation. This can also lead to the development of new research projects that address the specific needs and priorities of Indigenous communities.

The principles and recommendations outlined in this commentary will contribute to broader conservation and sustainability goals. By promoting a more inclusive and respectful approach to natural resource management, these strategies can help address global challenges such as biodiversity loss, climate change, and ecosystem degradation. The co-stewardship model can serve as a blueprint for achieving sustainable and equitable conservation outcomes, benefiting both human and non-human communities.

The implications of this commentary extend beyond moose management, offering valuable insights and frameworks that can be applied to various conservation efforts. By integrating Indigenous knowledge with scientific research and promoting inclusive governance, these recommendations can lead to more effective, equitable, and sustainable wildlife management practices.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

ETHICS STATEMENT

The authors declare that this synthesis adheres to the highest ethical standards. All information presented is based on verifiable sources and workshop outputs included in the supplemental material. The views expressed in this article are solely those of the authors and do not necessarily reflect the views of any affiliated institutions or organizations. No conflicts of interest exist that could influence the content or conclusions presented in this article.

DATA AVAILABILITY STATEMENT

Data available in article supplementary material

ORCID

Seth A. Moore http://orcid.org/0000-0002-5875-9306
William J. Severud http://orcid.org/0000-0003-0150-5986
Tiffany M. Wolf http://orcid.org/0000-0003-1740-4442

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